

A FEW CIRCUIT BLOCKS THAT ARE LIKELY TO BE PRESENT IN ANY AFE

Name of Circuit	Description	Use Case	Importance in AFEs
Trans-impedance Amplifier (TIA)	Commonly implemented on an op-amp, TIAs convert current to voltage and amplify the output to a suitable value.	They're used for sensors where the current response is more linear than the voltage response. This is true for photodiodes and optical sensors.	Since many applications like heart rate monitors employ LEDs and photodiodes, TIAs are essential. Depending on the sensors involved, different configurations of TIAs are used in AFEs. Apart from TIAs, many AFEs feature VGAs, PGAs, and many other kinds of amplifiers.
Filters	Filters remove unwanted frequency components from the applied signal and enhance the ones that are wanted.	A lot of times, signals coming from sensors are subjected to noise and interference, and hence these unwanted signals are removed using filters (high-pass, low-pass, or band-pass, depending on the sensor/application).	In AFEs, filters are used in many stages. Typically, they are used before passing a signal to an amplifier in the circuit. They are also used for anti-aliasing.
Sample and Hold (S&H) Circuits	These circuits are responsible for taking samples of the voltage given to them as input, and after that holding these samples for a definite time. This is usually done using an analogue switch, a capacitor, and a buffer.	They are generally used to improve the accuracy of ADCs. There are some ADCs that even have S&H circuits inbuilt.	Like in most applications, S&H circuits in AFEs are also present between filters and ADCs to maintain the accuracy of signals and to prevent errors due to amplitude change or skewing.
Analogue-to-Digital Converters (ADCs)	ADCs convert analogue signals into digital signals, usually in the two's complement binary form.	Most sensors like accelerometers, temperature sensors, flex sensors, etc are analogue sensors, and hence ADCs are used in most applications. Microcontroller development boards have ADCs onboard.	ADC is usually the final step in an AFE. The output of the ADC is given to the microcontroller or the next processing unit.
Instrumentation Amplifiers (InAmp)	InAmps are differential op-amps that provide high input impedances with gain adjustment to the heart rate through the variation of just one resistor.	They are useful in data-acquisition systems involving transducers and even RF signal systems.	In some AFEs, high-pass filters are coupled with instrumentation amplifiers so that large gain and high-pass filtering is possible in a single stage.

sometimes even application-specific integrated circuits (ASICs).

Heart rate monitoring techniques

There are several techniques for monitoring heart rate. All of these use different types of signals.

Phonocardiography (PCG). This technique uses the sound signal produced by the opening and closing of heart valves. As a rule, the signal is filtered and processed with the help of a digital signal processor (DSP). After proper filtering, it is possible

to detect not only heart rate but also abnormalities of the heart. This is done with the help of a machine called phonocardiograph.

Electrocardiography (ECG).

This technique is based on the measurement of the electric field produced by the heart. There must be two or more electrodes placed on specific points of the human body. Graphic representation of ECG consists of six peaks and valleys called P, Q, R, S, T, and U. The heart rate is measured by averaging RR interval between two R peaks (Fig. 1)

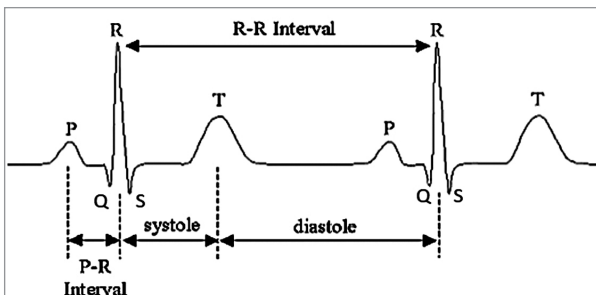


Fig. 1: Typical ECG of a healthy heart

over a defined period (for example, 15, 30, or 60 seconds).

Pressure sensing technique. This technique utilises a signal from a piezo sensor attached to one of the defined places on the human body, for example, on the wrist.

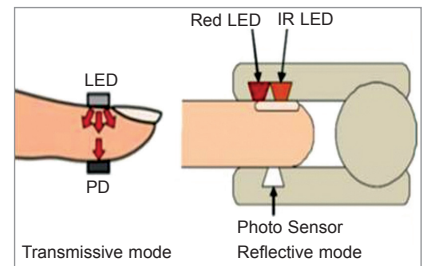


Fig. 2: The principle behind photoplethysmography (Credit: www.rfwireless-world.com)

Contraction and relaxation of the heart cause a corresponding change of pressure in the blood vessels, and therefore on the skin as well.

Photoplethysmography. This technique is based on the measurement of the volume variations of light reflected from the skin. When light emitted from an LED passes the heart through a blood vessel, the received light has pulsations that correspond to the heartbeat. Typically, a green or infrared LED is used as a light source because of the blood's tendency to absorb light with spe-